

Cross Product of Vectors in Matrix Form and Its Application to Electromagnetism

Ray Chien

September 19, 2026

1 The Cross Product Matrix

We know that in three-dimension space, the cross product of two vectors $\mathbf{A} \times \mathbf{B} = (A_x, A_y, A_z) \times (B_x, B_y, B_z)$ is defined as

$$(A_y B_z - A_z B_y, A_z B_x - A_x B_z, A_x B_y - A_y B_x).$$

In matrix form, it simply becomes

$$\begin{pmatrix} A_y B_z - A_z B_y \\ A_z B_x - A_x B_z \\ A_x B_y - A_y B_x \end{pmatrix} = \begin{pmatrix} 0 & -A_z & A_y \\ A_z & 0 & -A_x \\ -A_y & A_x & 0 \end{pmatrix} \times \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix}.$$

We define the **cross product matrix** $\mathbb{A}$ of a vector $\mathbf{A}$ as

$$\begin{pmatrix} 0 & -A_z & A_y \\ A_z & 0 & -A_x \\ -A_y & A_x & 0 \end{pmatrix}.$$

2 An Application

In electromagnetism, a particle with charge q with a velocity of $\mathbf{V}$ in space with electric field intensity $\mathbf{E}$ and magnetic flux density vector $\mathbf{B}$ should be subjected to the **Lorentz force** $\mathbf{F}$, which is given by

$$\mathbf{F} = q(\mathbf{E} + \mathbf{V} \times \mathbf{B}).$$

The question is, if we can give the particle arbitrary velocities and then measure the force acted on it. How many sets of experiments do we have to do?

To answer this, we first use our new invented cross product matrix to express the Lorentz force. Note that we now express vectors as column vectors.

$$\begin{aligned} \mathbf{F} &= q(\mathbf{E} + \mathbb{V}\mathbf{B}) \\ &= q \begin{pmatrix} I_3 & \mathbb{V} \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ \mathbf{B} \end{pmatrix} \end{aligned}$$

With two experiments, we have

$$\begin{pmatrix} \mathbf{F}_1 \\ \mathbf{F}_2 \end{pmatrix} = q \begin{pmatrix} I_3 & \mathbb{V}_1 \\ I_3 & \mathbb{V}_2 \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ \mathbf{B} \end{pmatrix}.$$

We hope $\begin{pmatrix} I_3 & \mathbb{V}_1 \\ I_3 & \mathbb{V}_2 \end{pmatrix}$ is invertible so that we can find $\mathbf{E}$ and $\mathbf{B}$. Unfortunately, the nullity of each cross product matrix of nonzero vector should be 1 because there is a null space, which lies in the direction of the vector. Therefore, we need at least three experiments with non-collinear velocities to obtain the electromagnetic field. For example, we have

$$\frac{1}{q} \begin{pmatrix} F_{1x} \\ F_{1y} \\ F_{2x} \\ F_{2y} \\ F_{3y} \\ F_{3z} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & -V_{1z} & V_{1y} \\ 0 & 1 & 0 & V_{1z} & 0 & -V_{1x} \\ 1 & 0 & 0 & 0 & -V_{2z} & V_{2y} \\ 0 & 1 & 0 & V_{2z} & 0 & -V_{2x} \\ 0 & 1 & 0 & V_{3z} & 0 & -V_{3x} \\ 0 & 0 & 1 & -V_{3y} & V_{3x} & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \\ B_x \\ B_y \\ B_z \end{pmatrix} = M \begin{pmatrix} E_x \\ E_y \\ E_z \\ B_x \\ B_y \\ B_z \end{pmatrix},$$

and linear algebra tells us that M invertible. Ask [Zhenverse](#) if you have questions about linear algebra.